

OEM Power Converter Module: VC-PSU-480W

Integrated 24V DC Internal Auxiliary Power Supply for Motion Systems

1. Technical Specifications Summary

Parameter	Rated Specification	Notes / Derating
Nominal Output Voltage	24.0 V DC $\pm 1.0\%$	Factory calibrated
Maximum Output Current	20.0 A continuous	Peak 25.0 A for 3s
Output Voltage Ripple & Noise	120 mV pk-pk maximum	Measured at 20 MHz
Operating Ambient Temperature	-10.0 °C to +40.0 °C	Derate 2.5%/°C above 40°C
Storage Temperature	-40.0 °C to +85.0 °C	Standard non-condensing
Isolation Voltage (Input/Output)	3.0 kV AC RMS	1 minute qualification

2. Engineering Derating Advisory & Thermal Notes

CRITICAL APPLICATION NOTE: The VC-PSU-480W auxiliary power converter is rated for nominal full-load operation up to +40.0 °C ambient temperature. When operated in enclosed environments exceeding +40.0 °C, the module enters thermal derating mode (output current limited to 12.0 A at +50.0 °C). Full operation at +55.0 °C is NOT supported without supplemental forced-air cooling (minimum 2.5 m/s airflow).

Output Ripple: Under maximum load, internal switching ripple on the secondary rail is guaranteed below 120.0 mV pk-pk. Applications requiring < 50 mV pk-pk ripple must install an external LC low-pass filter stage.